

Milestone 10: Testing, Deployment, and Project Closure

Objective

The objective of this phase was to validate the functionality, security, performance, and usability of the Smart Leave Approval System before deployment. This phase ensured that all project requirements were successfully implemented and that the application was ready for production use.

Overview

Testing and deployment are critical phases in the software development lifecycle. During this phase, the application was thoroughly tested to verify that all modules, workflows, AI features, dashboards, and security configurations functioned as expected.

After successful testing, the application was prepared for deployment and final project review.

Testing Strategy

The testing process focused on:

1. Functional Testing
 2. Security Testing
 3. Workflow Testing
 4. Dashboard Testing
 5. User Acceptance Testing (UAT)
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Functional Testing

Test Case 1: Apply Leave Request

Objective

Verify that employees can successfully submit leave requests.

Steps

1. Login as Employee.
2. Open Apply Leave module.
3. Enter leave details.
4. Submit request.

Expected Result

A new leave request record is created.

Actual Result

Request submitted successfully.

Status

 Passed

Test Case 2: Total Days Calculation

Objective

Verify automatic calculation of leave duration.

Input

Start Date	End Date
10-Jun-2026	12-Jun-2026

Expected Result

Total Days = 3

Actual Result

Calculated correctly.

Status

 Passed

Test Case 3: AI Risk Assessment

Objective

Verify AI-based risk classification.

Input

Leave Duration = 8 Days

Expected Result

High Risk

Actual Result

Risk generated correctly.

Status

 Passed

Test Case 4: AI Recommendation

Objective

Verify AI recommendation generation.

Expected Result

Extended leave requires additional review.

Actual Result

Recommendation generated successfully.

Status

 Passed

Workflow Testing

Test Case 5: Approval Workflow

Objective

Verify leave approval process.

Steps

1. Employee submits request.
2. Manager reviews request.
3. Manager approves request.

Expected Result

Status changes to:

Approved

Actual Result

Workflow executed successfully.

Status

 Passed

Test Case 6: Rejection Workflow

Objective

Verify rejection process.

Steps

1. Employee submits request.
2. Manager rejects request.

Expected Result

Status changes to:

Rejected

Actual Result

Workflow executed successfully.

Status

 Passed

Security Testing

Test Case 7: Employee Access

Objective

Verify employee permissions.

Expected Result

Employee can:

- Apply Leave
- View Own Requests

Employee cannot:

- Access Dashboard
- Approve Requests

Result

Security controls functioned correctly.

Status

 Passed

Test Case 8: Manager Access

Objective

Verify manager permissions.

Expected Result

Manager can:

- View Requests
- Approve Requests
- Reject Requests
- Access Dashboard

Result

Permissions configured correctly.

Status

 Passed

Dashboard Testing

Test Case 9: Dashboard Functionality

Objective

Verify dashboard visualization.

Expected Result

Dashboard displays:

- Total Leave Requests
- Requests by Status
- Requests by Leave Type
- AI Risk Analysis

Result

Dashboard loaded successfully.

Status

 Passed

User Acceptance Testing (UAT)

Objective

Validate application usability and functionality from an end-user perspective.

Employee Feedback

- Easy to submit leave requests.

- Clear approval status visibility.
- User-friendly interface.

Manager Feedback

- Approval process simplified.
- AI recommendations helpful.
- Dashboard provides useful insights.

Outcome

User acceptance criteria successfully met.

Deployment Activities

After successful testing, the application was prepared for deployment.

Deployment Checklist

- ✓ Application Validated
 - ✓ Business Rules Tested
 - ✓ ACLs Verified
 - ✓ Flow Designer Workflow Tested
 - ✓ Reports Verified
 - ✓ Dashboard Verified
 - ✓ User Roles Assigned
 - ✓ Sample Data Validated
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Project Achievements

The Smart Leave Approval System successfully achieved the following objectives:

Automation

- Automated leave request processing.

- Reduced manual effort.

AI Decision Support

- Risk assessment implementation.
- Recommendation generation.

Security

- Role-based access control.
- Secure data management.

Analytics

- Dashboard implementation.
- Reporting and KPI monitoring.

Challenges Faced

During development, several challenges were encountered:

Form Visibility Issues

Employee users could not initially view form fields due to ACL configuration issues.

Dashboard Access

Role-based restrictions required additional dashboard configuration.

AI Risk Assessment Logic

Business rules required refinement to ensure proper risk classification.

Resolution

All issues were successfully identified, analyzed, and resolved during testing.

Future Enhancements

The following enhancements can be implemented in future versions:

- Virtual Agent Integration
 - Agentic AI Features
 - Leave Balance Management
 - Email Notifications
 - Mobile Application Support
 - Predictive Leave Analytics
 - Employee Self-Service Portal
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Conclusion

The Smart Leave Approval System was successfully developed and deployed on the ServiceNow platform. The application automates leave request submission, approval processing, AI-based risk assessment, security management, and reporting.

The system improves operational efficiency, enhances transparency, supports managerial decision-making, and provides a scalable solution for leave management within an organization.
